



EXAMEN DE FIN D'ÉTUDES SECONDAIRES – Sessions 2025

QUESTIONNAIRE

Date :	22.05.25	Horaire :	08:15 - 11:15	Durée :	180 minutes	
Discipline :	AMINF	Type :	écrit	Section(s) :	CI	
					Numéro du candidat :	

Available documents:

- Answers.dotx: template to be used to insert your answers into.
- Translation.mod: contains the CDM that is to be translated.
- Database.sql: MySQL database with a sample dataset.
- Functions.pdf: list of known SQL functions.

Preparation and hand-in

In your repository (to be defined by each school), you will find a folder named **EXAMEN_AMINF**. Rename this folder by replacing its current name with your exam code (example: **LXY_CI_07** where LXY is to match your school). All your files must be saved inside this folder, which will be referred to as **your folder** in the remainder of this document.

Next, open the file **Answers.dotx** which resides in your folder. Fill in the header of this file by adding your candidate number and the current date. Save the document inside your folder (example notation: **LXY_CI_07_Answers.docx**).

At the end of the exam, create a PDF version of your answer file (example notation: **LXY_CI_07_Answers.pdf**) and ensure that:

- The PDF file contains all your answers.
- The screenshots are legible.
- All the SQL queries are provided, one query per file.

Remember to save your document regularly!

You will be evaluated solely on the contents of the PDF file!

Question 1 – Establishing a CDM**[22p]**

Create in Modelizer a *Conceptual Data Model* (CDM) for a mobile application named *HelpShare* based on the information and the mock-up from the following description. Upon completion, include a screenshot in your answer document.



The software to be modelled is a community-driven app where people can request help for small tasks, such as changing a tyre, mowing the lawn, or help with groceries. These tasks are meant to be done for free, but the person requesting help can leave a tip as a token of appreciation.

To use the app, users must first create an account. During registration, the username, password, e-mail address, first name, last name, and a home address must be provided. The verification status (verified, pending, unverified), along with information on whether the user is a staff member, is also stored to help the company better manage its users.

Users can verify their identity at any time to access all of the app's features. Until then, they are limited to browsing the app. To verify their identity, users simply upload an identification document. The expiration date and ID number are automatically scanned and stored, along with the full file path (including the file name) of the scanned document on the server. Supported identification documents have a type (passport, ID card, driving permit, etc) and a minimal age requirement per type to ensure only eligible options are displayed based on the user's age. The verification date is also recorded, indicating when a user's document was verified, if at all.

Each new user is assigned another, more experienced user—a mentor—who provides guidance during their initial phase. Over time, mentors gain experience by assisting multiple users. Staff members, do not go through this mentorship process.

Once a user is verified, they can begin offering to complete tasks for others or apply to take on tasks posted by others. Tasks may belong to multiple categories, such as Kitchen, Vehicle, Outdoor, etc., or remain uncategorized. Each category includes a detailed description that outlines what kind of tasks fall under that category, ensuring clarity for users. Each task must include a summary and may include a detailed description. The optional start and end dates for the task are also stored. A task may include a location to inform applicants of where the task is to be carried out. This can range from a precise address to a broader reference, such as a city name, enabling applicants to gauge the distance involved. Each task has a status indicating its progress, such as open or closed. Users may mark tasks to watch, allowing for quicker and easier access without the need to search for them again.

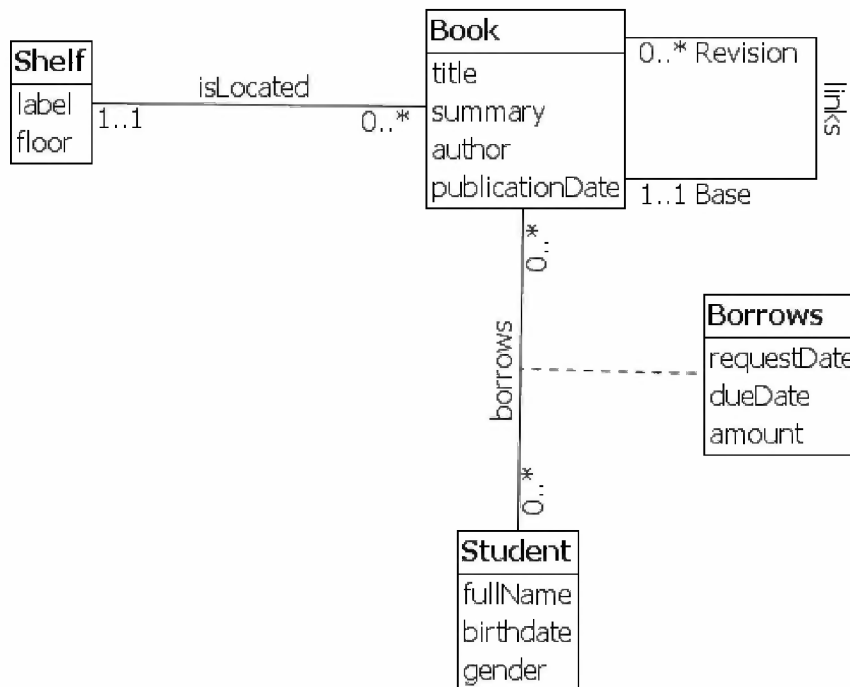


Figure 1: A mock-up of the app

Tasks may include multiple images uploaded in a specific order. Each image may include a comment that will be visible to the applicants in the app. The full file path of the image is also stored. Uploaded images for a task cannot be used in other tasks. When a task is completed, an optional tip amount can be entered for statistical purposes. Any verified user can apply for a pending task and the task initiator may accept multiple applicants. Whenever a user applies for a task, the submission date, its current status (accepted, rejected, pending), and an optional review text of the overall experience are recorded.

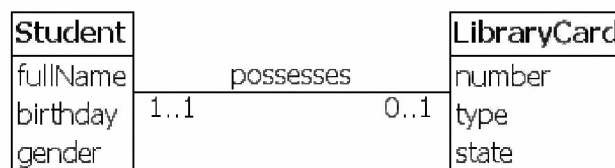
Question 2 – LDM and Comprehension**[15p]**

- a) Open the Translation.mod file in Modelizer. Translate the CDM below into a *Logical Data Model* (LDM). Adhere to the conventions covered. Insert a screenshot of your LDM into your document. **[9p]**



Note: It is not possible to borrow a set of books more than once per day.

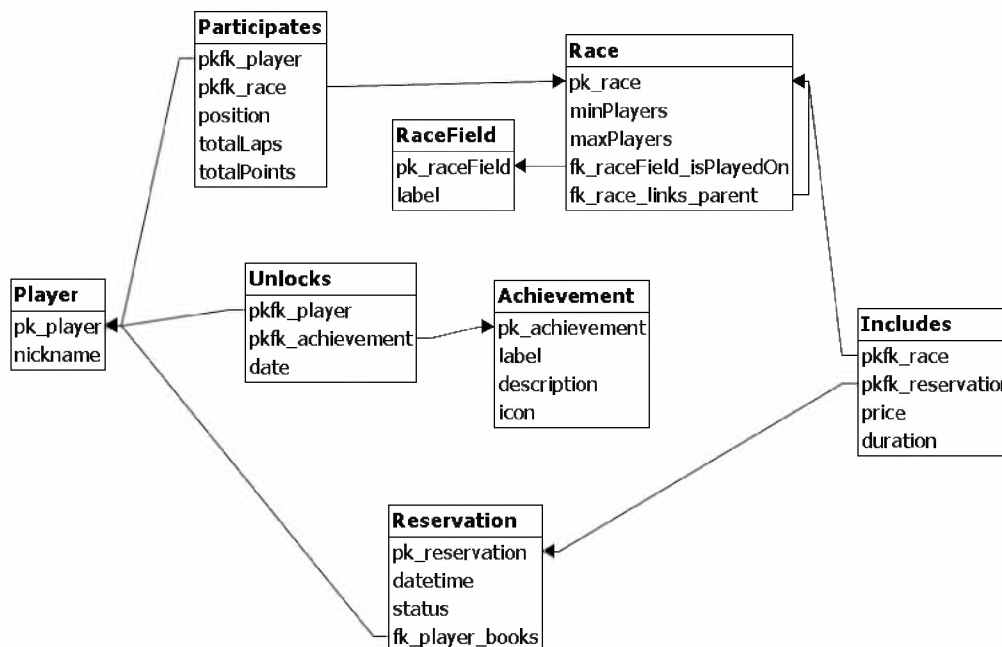
- b) Consider the following extension:



When translating the CDM into an LDM, the foreign key may be placed in the Student or LibraryCard table. Formulate an argument in favour or against each of these choices. **[6p]**

Question 3 – SQL Queries and Comprehension**[23p]**

Import the Database.sql file into MySQL Workbench to create a database preloaded with test data corresponding to the LDM below. You may need to add additional entries to verify your queries are correct.



Q1. Retrieve the alphabetically ordered nicknames of players who have never made a reservation.

[3p]

Q2. Provide the SQL code required to insert a new record into the Achievement table. You may choose any value(s) that comply with the table's definition.

[2p]

Q3. Formulate the following SQL query in a sentence.

[2p]

```

1 • SELECT
2     nickname
3 FROM
4     player
5 WHERE
6     nickname NOT LIKE "% %";
    
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Q4. Display all races that could not be held because of insufficient participants. Ensure that the column names and their order match the format depicted below.

[7p]

Race id.	Race Field	Minimum Players	# Participants
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Q5. Retrieve the top 3 ranked nicknames based on their accumulated points in all races. **[4p]**

Q6. Develop the SQL query to display the following result based on the initial data set: **[5p]**

Achievement	Players
Top Scorer	KriRo,PirSy
Lap Leader	TobEr,ThiFr
Gold	HOLL
Fast Starter	GasDr,GreBj
Endurance Racer	HaaLa,WebRe